



PROFILE

Graduate researcher with **3+ years of experience** in Natural Language Processing and Retrieval-Augmented Generation (RAG). Co-author of a multimodal agentic RAG publication (ColLEX). Expert in Python-based AI backends, with a proven ability to bridge research models into functional prototypes using Flask and collaborative integration with Django and Next.js.

CONTACT DETAILS

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LANGUAGES

English (C1 / IELTS), German (B1), Persian (native)

SKILLS

- Python, C/C++, SQL, JavaScript
- PyTorch, scikit-learn, pandas, NumPy
- Flask, Django, Next.js, Chroma, SQLAlchemy
- Docker, Git, $\LaTeX$, OpenWebUI

CERTIFICATES

- ◇ Advanced Learning Algorithms
- ◇ Supervised Machine Learning
- ◇ IELTS (British Council, Oct 2023)

EDUCATION

- M.SC. INTELLIGENT ADAPTIVE SYSTEMS **2024.10–pres.**
◇ **Universität Hamburg**
- B.SC. COMPUTER SCIENCE **2019.09–2024.02**
◇ **University of Tehran**

EXPERIENCE

Research Assistant / Intern at *University of Hamburg* **2024.11–2026.04**
◇ **POL-App (Medical RAG):** Developed a diagnostic support system. Bridged AI backends with user interfaces by implementing chat streaming and multi-lingual behavior using **Django** and **Next.js**. **2026.03–2026.04.**

◇ **IMV Lab Assistant:** Designed a domain-specific RAG system for SDG indicators and impact reporting. Optimized retrieval using **sentence-transformers** and structured German-language prompting. **2025.11–2026.02**

◇ **UHH Tool Catalogue:** Built a conversational tool discovery system using **Flask**, **SQLAlchemy**, and **Chroma**. Integrated LLMs via **OpenWebUI** and containerized for deployment with **Docker**. **2025.03–2025.09**

◇ **Software Engineering RAG:** Led LLM integration and designed a typed function-calling interface. Migrated stack from OpenAI to **Google Gemini** and improved robustness through unit testing. **2024.11–2025.03**

Research Intern at *University of Hamburg*. **2023.07–2024.02**

◇ Investigated few-shot and zero-shot LLM capabilities for word-sense disambiguation (WSD). Evaluated prompt-tuning vs. fine-tuning and developed an ontology-based classification prototype.

Visiting Scholar at *KU Leuven (Leuven, Belgium)*. **2022.07–2022.09**

◇ Analyzed Speech Recognition calibration; implemented an MLP-based approach to improve character-level confidence estimation.

PUBLICATIONS

COLLEX: A Multimodal Agentic RAG System Enabling Interactive Exploration of Scientific Collections 2025

◇ F. Schneider, N. Baba Ahmadi, **Niloufar Baba Ahmadi**, I. Vogel, M. Semmann, C. Biemann [[arXiv:2504.07643](https://arxiv.org/abs/2504.07643)]

- Co-developed an agentic RAG system for exploring 64,000+ scientific records.
- Implemented the typed function-calling interface and multimodal retrieval orchestration logic.

TEACHING & VOLUNTEERING

Chief Teaching Assistant at *University of Tehran*. **2021.09–2024.03**

◇ **Advanced Information Retrieval:** Coordinated a team of assistants, designed student projects, and supported teaching for Prof. Bagher BabaAli.

◇ **Fundamentals of CS & Programming:** Served as Chief TA under Prof. Mousavian and Dr. Ebrahimnejad; managed course instruction support and grading workflows for multiple cohorts.

OE-Tutor & PIASTA Welcome Buddy at *University of Hamburg*. **2025.05–2025.10**

◇ Guided international students through academic orientation and cultural integration within the university.

THESIS PROJECT

Bachelor's Thesis: EEG-Based Classification of Mental Disorders at *University of Tehran*. **2023.04–2023.07**

- ◇ Developed and evaluated a machine-learning pipeline for classifying mental-disorder categories from EEG-derived features and clinical/demographic variables.
- ◇ Conducted model selection and hyperparameter optimisation (ElasticNet logistic regression, SVM, Random Forest) using cross-validation and systematic comparisons.
- ◇ Results: achieved **0.9889** accuracy for major-disorder classification (SVM) and **0.8104** for specific/minor disorders (ElasticNet) on a dataset of 945 samples.

PROJECTS

FUNDus MuRAG (Publication project) *RAG, OpenAI API, Python, Docker*

[Repository](#)

- ◇ Integrated OpenAI models into an existing multimodal RAG system, adapting prompts and generation logic for grounded Q&A. Contributed to a 2025 workshop publication.

Parkinson Disease Classification *Python, pandas, NumPy, scikit-learn*

[Repository](#)

- ◇ Built a classification workflow on biomedical voice features with preprocessing, PCA-based dimensionality reduction, and model evaluation.

House Price Prediction *Python, pandas, NumPy, scikit-learn*

[Repository](#)

- ◇ Developed a supervised regression pipeline with preprocessing and baseline comparison (linear and tree-based models).

Maze Navigation using Q-Learning *Python, NumPy, networkx, matplotlib*

[Repository](#)

- ◇ Implemented Q-learning for agent navigation in a grid world and visualised learning across reproducible training runs.